1.7 Feynman diagrams, Dyson equation, Self-energy

How to perform calculations? One of the most natural way is to perform perturbation theory for the Green function, whose coefficients can be systematically evaluated in the unified formalism of Feynman diagrams.

Perturbation theory expansion of G

The expansion is done ~~in~~ in the electrons interaction V_I

$$H = \underbrace{\sum_{k\sigma} \varepsilon_{\vec{k}} c^{+}_{\vec{k}\sigma} c_{\vec{k}\sigma}}_{H_0} + \frac{1}{2} \sum_{\substack{\vec{k}\vec{k}'q\\ \sigma\sigma'}} \underbrace{V(q)\, c^{+}_{\vec{k}+\vec{q},\sigma} c^{+}_{\vec{k}'-\vec{q},\sigma'} c_{\vec{k}',\sigma'} c_{\vec{k},\sigma}}_{V_I}$$

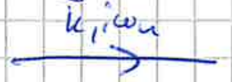
interaction (Dirac) picture:
$$\tilde{c}_k(\tau) = e^{H_0\tau} c_k e^{-H_0\tau}$$
$$\hat{V}_I(\tau) = e^{H_0\tau} V_I e^{-H_0\tau}$$

$\Rightarrow$ allows to perform an expansion in powers of $\hat{V}_I(\tau)$, whose coefficients are depending only on H_0

$$G(\vec{k},\tau) = -\frac{1}{Z} \langle T_\tau \tilde{c}_{\vec{k}}(\tau) \tilde{c}^{+}_{\vec{k}}(0) \Big[\mathbb{1} - \int_0^\beta d\tau_1 \hat{V}_I(\tau_1) + \dots$$
$$\dots + \frac{(-1)^n}{n!} \int_0^\beta d\tau_1 \dots d\tau_n \tilde{V}_I(\tau_1) \dots \tilde{V}(\tau_n) + \dots \Big] \rangle_0$$

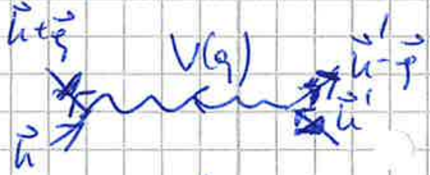
(Z also has to be expanded)

These expansions can be done diagrammatically as follows:

Main ingredients: i) 

free propagation

$$G_0 = \frac{1}{i\omega_n - \xi_k}$$

ii) 

Coulomb interaction

$$V(q) = \frac{e^2}{\varepsilon_0 q^2}$$

(*)

They are combined by the Feynman rules

(*) R. Feynman (1918-1988)

Step 1: Drawing $\Rightarrow$ draw all topologically non-equivalent and connected diagrams with n interaction and $(2n+1)$ bare propagation lines with energy / momentum conservation at each vertex.

Step 2: Evaluating $\Rightarrow$ include a factor $V(q)$ for each interaction line and a factor of $G_0 = \frac{1}{i\omega_n - \xi_k}$ for each propagator line. Integrate over all internal degrees of freedom (e.g. $\frac{1}{(2\pi)^d}\int d^d k \frac{1}{\beta}\sum_{\omega_n} \ldots$) and multiply an overall prefactor

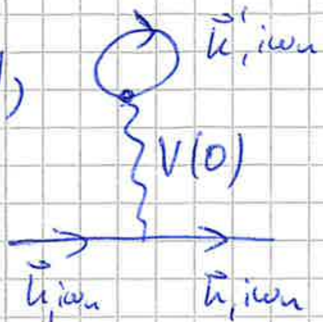
$(-1)^n (-1)^F (2)^F$ with n ... expansion order

F ... # of closed loops in diagram

Examples for the lowest order ($n=1$)

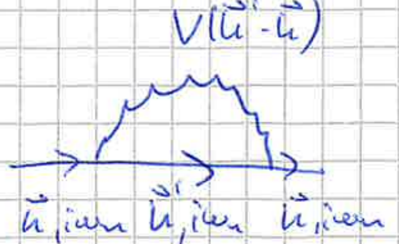
Step 1: Drawing: with $n = 1$ and $2n+1 = 3 \rightarrow$ lines we can draw two topologically distinct and connected diagrams:

1)



Hartree diagram

2)



Fock diagram

All other diagrams are either disconnected ()

or equivalent.

Step 2: Evaluating

example: Hartree diagram

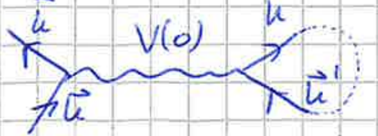
$$= \underbrace{(-1)^1}_{n} \underbrace{(-1)^1 (2)^1}_{F} \left[G_0(\vec{k}, i\omega_n) \right]^2 V(0) \cdot$$

$$\cdot \frac{1}{(2\pi)^d} \int d^d k' \, \frac{1}{\beta} \sum_{n'} G_0(\vec{k}', i\omega_{n'}) =$$

$$= 2 \left[G_0(\vec{k}, i\omega_n) \right]^2 V(0) \, G_0(\vec{r}=0, \tau=0)$$

$\Downarrow$ ambiguous $\rightarrow$ G_0 has jump at $\tau = 0$!

look at $\hat{V}_t = c^\dagger c^\dagger c c$



$\Rightarrow$ we have to swap order $\Rightarrow \tau = 0^-$ $\quad \Rightarrow G_0(\tau) = -\langle c(\tau) c^\dagger(0) \rangle$

$$G_0(\tau = 0^-) = \langle c^\dagger(0) c(0) \rangle =$$

$$= \langle n_0 \rangle = \underset{\text{density}}{n_0}$$

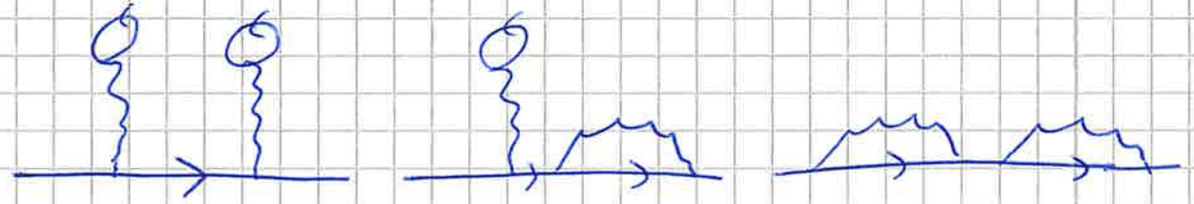
$$\Rightarrow \; = 2 \left[G_0(\vec{k}, i\omega_n) \right]^2 V(0) \, n_0(\vec{r}=0) =$$

$$= \frac{2 V(0) \, n_0(\vec{r}=0)}{(i\omega_n - \varepsilon_{\vec{k}})^2}$$

Higher order diagrams

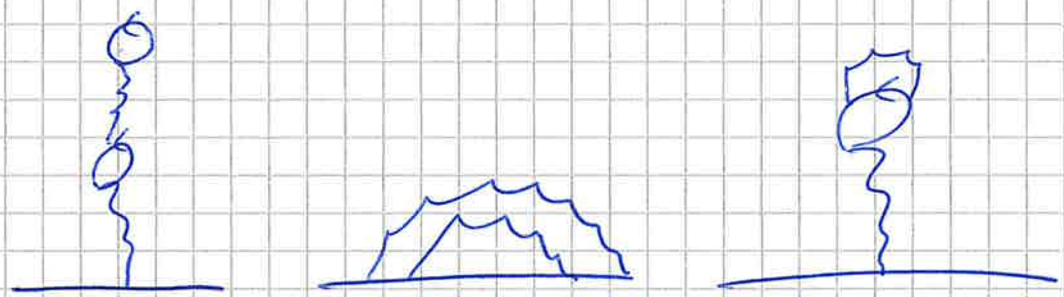
Already for $n=2$ there are many more and can be classified into:

type A:



diagrams, where a first-order process is repeated n times

type B:



diagrams, where a first order process "corrects" / "dresses" the internal lines

type C:

diagrams of genuine second order nature.

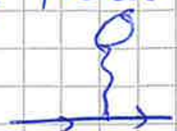
As only diagrams of type C describe completely new processes w.r.t. the first order (Hartree and Fock), a systematic way to include all diagrams of types A and B is desirable

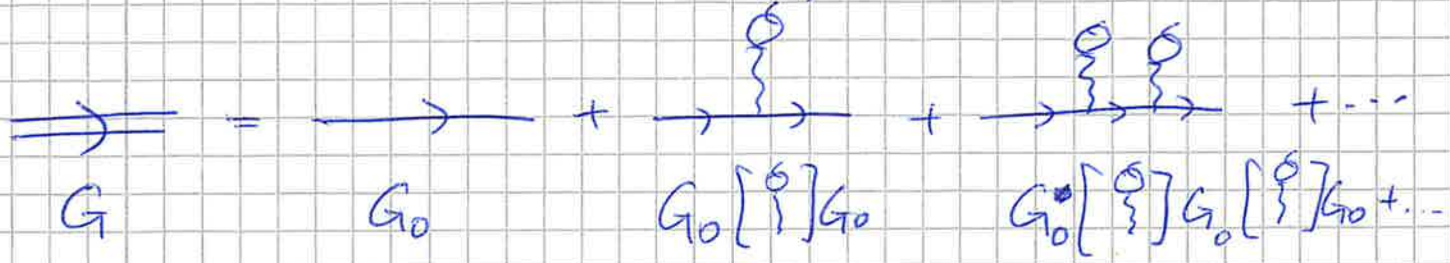
type A $\rightarrow$ Dyson equation (self-energy)

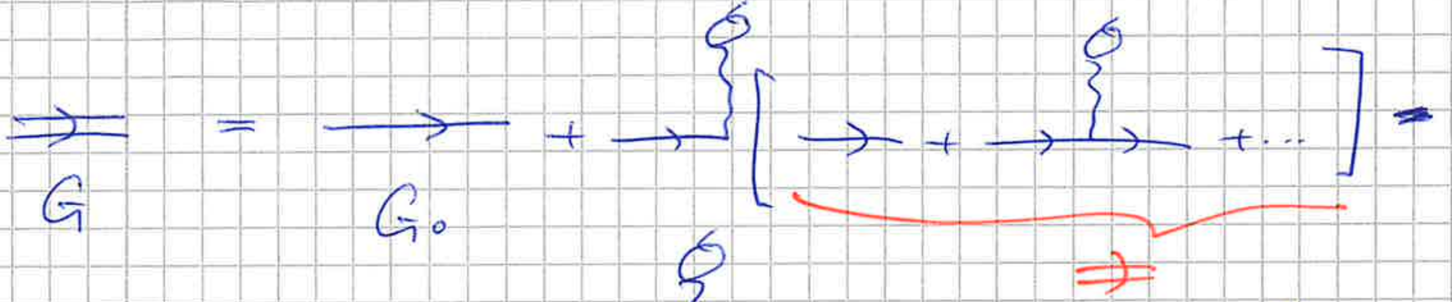
type B $\rightarrow$ self-consistent ρ

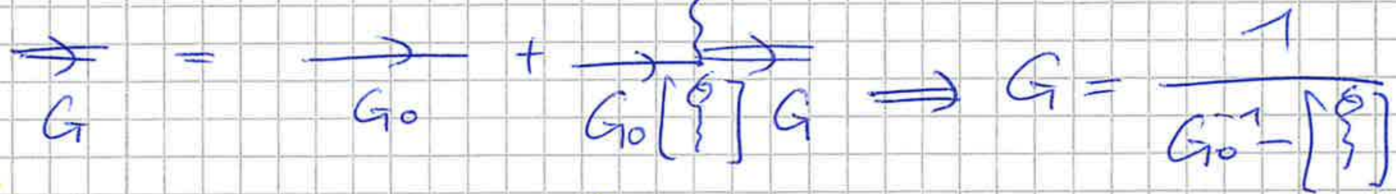
Dyson equation and Self-energy

Diagrams of type A → the same process is repeated $n = 1, 2, \ldots$ times ⇒ summing ALL of them!

Let us consider the Hartree diagram  and define







This can be done also for other diagrams. The important point is <u>not</u> to include diagrams, which are generated by the series (avoid double counting)

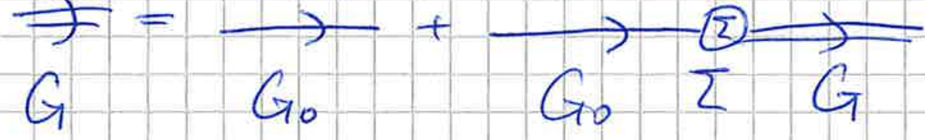
⇒ definition of the <u>self-energy</u> Σ

 : all topologically distinct diagrams, which cannot be split into two parts by cutting one propagator (⇒ irreducible diagram)

e.g.: all type A diagrams are reducible

 ⇒ do not belong in 

After choosing a set of diagrams among type B and C:



$\Downarrow$

Dyson equation

$$\boxed{G(\vec{k}, i\omega_n) = \frac{1}{G_0^{-1}(\vec{k}, i\omega_n) - \Sigma(\vec{k}, i\omega_n)} = \frac{1}{i\omega_n - \varepsilon_{\vec{k}} - \Sigma(\vec{k}, i\omega_n)}}$$

This does not only allow for a systematic treatment of all diagrams of type A (reducible), but also provides a much more physical interpretation of the interaction corrections
$\Rightarrow$ approximation derived for Σ than G

$$G = \frac{1}{i\omega_n - \varepsilon_{\vec{k}} - \underbrace{\mathrm{Re}\,\Sigma(\vec{k}, i\omega_n)}_{\to 0} - i\,\underbrace{\mathrm{Im}\,\Sigma(\vec{k}, i\omega_n)}_{\to 0}} \xrightarrow[\mu\,\Sigma]{i\omega_n \to 0} \frac{1}{i\omega_n - \varepsilon_{\vec{k}} - \mathrm{Re}\,\Sigma(\vec{k}) - i\,\mathrm{Im}\,\Sigma(\vec{k})}$$

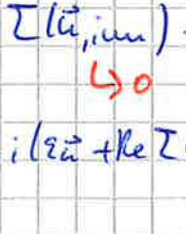
limits

correction to $\varepsilon_{\vec{k}}$

finite lifetime τ: γ

$$\Rightarrow G(t) \approx e^{-i(\varepsilon_{\vec{k}} + \mathrm{Re}\,\Sigma(\vec{k}))t}\, e^{-\gamma t} \to \text{decay}$$

type B: Self-consistency

$\Sigma =$  $\rightarrow$ 

$$G = \frac{1}{i\omega_n - \varepsilon_{\vec{k}} - \Sigma_{\text{Hartree}}}$$ instead of G_0

$\Rightarrow$ iteratively

Example: Self-energy of a Fermi liquid

Most of the properties of metallic systems can be qualitatively understood even with non-interacting calculations (e.g.: $C_V(T) \propto T$, $\chi_m(T) \propto \text{const}, \ldots$), just the values of prefactors are different ($\to$ renormalization)

$\Rightarrow$ Structure of electronic excitations for $E \approx E_{Fermi}$ resembles non-interacting one ($\Rightarrow$ Fermi liquid theory)

microscopic point of view: $-\Sigma-$ can be Taylor-expanded around the Fermi level ($|\vec{k}| \to k_F$, $i\omega_n \to 0$)

$$\Sigma(\vec{k}, i\omega_n) \cong \underbrace{\mathrm{Re}\,\Sigma(k_F, i\omega_n \to 0)}_{\text{reabsorbed in chem. pot. } \mu} + \underbrace{\frac{\partial \mathrm{Re}\,\Sigma(k_F, 0)}{\partial k}}_{\text{correction to } \varepsilon_k}(k - k_F) +$$

$$+ \underbrace{i\,\mathrm{Im}\,\Sigma(k_F, 0)}_{-i\Gamma} + \underbrace{i\,\frac{\partial\,\mathrm{Im}\,\Sigma(k_F, 0)}{\partial \omega_n}}_{-i\alpha\omega_n} + \ldots$$

$$G(\vec{k}, i\omega_n) = \frac{1}{i\omega_n - \varepsilon_{\vec{k}} - \Sigma(\vec{k}, i\omega_n)} \overset{\text{for } \omega_n \to 0}{=} \frac{1}{i\omega_n(1+\alpha) - \Big(\underbrace{\frac{\partial \varepsilon_k}{\partial k}}_{v_F^0} + \frac{\partial \Sigma}{\partial k}\Big)_{k_F}(k - k_F) + i\Gamma + \ldots}$$

$$Z = (1+\alpha)^{-1} < 1$$

$$= \frac{1}{\frac{i\omega_n}{Z} - \left(v_F^0 + \frac{\partial \Sigma}{\partial k}\right)(k - k_F) + i\Gamma} = \frac{Z}{i\omega_n - Z\left(v_F^0 + \frac{\partial \Sigma}{\partial k}\right)_{k_F}(k - k_F) + iZ\Gamma}$$

Summary:

- non-interacting system $\Rightarrow$ particle excitations with disp. $v_F^0(k-k_F)$; $\tau=\infty$ infinite lifetime
- Fermi liquid (FL) $\Rightarrow$ QUASI-particle excitations with disp. $Z\left(v_F^0+\frac{\partial\Sigma}{\partial k}\right)(k-k_F)$ and weight $Z<1$; $\tau=\frac{\hbar}{\Gamma}<\infty$

Looking at plot of $\mathrm{Im}\,\Sigma(k_F, i\omega_n)$: